

Non-Minimum-Phase Channel DoA Estimation via UCA Phase-Mode Smoothing and Enhanced Subspace Separation

Bachelor's Thesis

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1 Introduction

Eigenstructure-based DoA estimation relies on correctly separating the signal and noise subspaces of the data covariance matrix. In practical combat-surveillance deployments, however, ideal channel assumptions rarely hold. We consider a particularly adverse scenario in which the desired transmission is spoofed by a higher-power signal arriving from a different direction, while also accounting for real-world impairments such as multipath propagation. Under these conditions, conventional model-order selection criteria fail in the presence of coherent sources and multipath. To address this, we introduce a dynamic thresholding scheme and derive the corresponding mathematical threshold condition for subspace separation. We further analyze the Cramér–Rao bound and show that the proposed method approaches this bound at sufficient SNR.

2 Background

An incoming plane wave from direction $\hat{r}$ is modeled at the n^{th} array element as

$$x_n(t) = \tilde{g}_0(t), e^{jK_0\hat{r}\cdot\vec{P}_n}, \quad (1)$$

where $\tilde{g}_0(t) = Ae^{j\alpha}$ is the complex envelope at the origin, $\vec{P}_n$ is the position vector of the n^{th} element, $K_0 = \frac{2\pi}{\lambda}$ is the wavenumber, and $\hat{r}$ is the DoA unit vector.

The steering vector of an N -element array is

$$\underline{\mathbf{a}}(\hat{r}) = \begin{bmatrix} e^{jK_0\hat{r}\cdot\vec{P}_0} \\ e^{jK_0\hat{r}\cdot\vec{P}_1} \\ \vdots \\ e^{jK_0\hat{r}\cdot\vec{P}_{N-1}} \end{bmatrix} \quad (2)$$

For an UCA of radius r in the $\theta = 90^\circ$ plane (XY-plane), the position vector of the n^{th} element is:

$$\vec{P}_n = \begin{bmatrix} r \cos \gamma_n \\ r \sin \gamma_n \\ 0 \end{bmatrix}, \quad \text{where } \gamma_n = \frac{2\pi(n-1)}{N} \quad (3)$$

Using complex envelope notation, the UCA steering vector [1] becomes:

$$\mathbf{a}(\theta, \phi) = \begin{bmatrix} e^{j\zeta \cos(\phi-\gamma_0)} \\ e^{j\zeta \cos(\phi-\gamma_1)} \\ \vdots \\ e^{j\zeta \cos(\phi-\gamma_{N-1})} \end{bmatrix}, \quad \text{where } \zeta = K_0 r \sin \theta \quad (4)$$

The array data model is:

$$\mathbf{X}(t) = \mathbf{A} \cdot \mathbf{S}(t) + \mathbf{n}(t) \quad (5)$$

where $\mathbf{S}(t)$ is the signal vector at the reference point, and $\mathbf{n}(t)$ is the noise.

Instead of weighting individual UCA elements arbitrarily, the weights are defined as a function of each element's angular position:

$$w(\gamma_n) = e^{jm\gamma_n}, \quad \text{where } \gamma_n = \frac{2\pi(n-1)}{N} \quad (6)$$

m is the **mode**, an integer defining the spatial “harmonic” [2]. These **spatial harmonics**, just like in Fourier series, can form arbitrary patterns by superposition:

$$w(\gamma) = \sum_{m=-M}^M c_m e^{jm\gamma} \quad (7)$$

Applying mode- m weights to the UCA element outputs and summing gives the array output:

$$y_m(t) = \sum_{n=0}^{N-1} x_n(t) \cdot e^{jm\gamma_n} \quad (8)$$

This operation maps the array from element space to beamspace. The maximum usable mode order is $M = \lfloor K_0 r \rfloor$, where r is the radius of the circular array.

The far-field array pattern for mode m ($N > 2M$, which is the equivalent of the Nyquist criterion [3]) becomes:

$$A_m(\phi) = j^m \cdot J_m(\zeta) \cdot e^{jm\phi} \quad (9)$$

where $J_m(\zeta)$ is the Bessel function of the first kind (order m) and $\zeta = K_0 r \sin \theta$, and ϕ is the azimuth angle. The exponential term induces a Vandermonde-like structure analogous to ULAs.

The beamformer matrix $\mathbf{F}$ satisfies $\mathbf{F}\mathbf{F}^H = \mathbf{I}$ to preserve white-noise statistics. The matrix of modal weights is defined as:

$$\mathbf{V} = \sqrt{N} [\mathbf{w}_{-M} \quad \cdots \quad \mathbf{w}_0 \quad \cdots \quad \mathbf{w}_M] \quad (10)$$

$$\mathbf{w}_m = \frac{1}{N} [e^{jm\gamma_0}, \dots, e^{jm\gamma_{N-1}}]^T \quad (11)$$

Its Hermitian $\mathbf{V}^H \in \mathbb{C}^{(2M+1) \times N}$ has each row as a beamformer for a spatial mode. The **even** (complex) beamformer matrix is defined as:

$$\mathbf{F}_e^H = \mathbf{C}_e \mathbf{V}^H, \quad \mathbf{C}_e = \text{diag}(j^{-M}, \dots, j^{-1}, 1, j^1, \dots, j^M) \quad (12)$$

To obtain a real-valued beamspace representation,

$$\mathbf{F}_r^H = \mathbf{W}^H \mathbf{F}_e^H \quad (13)$$

$$\mathbf{W}^H = \frac{1}{\sqrt{M'}} \begin{bmatrix} \mathbf{v}(\alpha_{-M})^H \\ \vdots \\ \mathbf{v}(\alpha_0)^H \\ \vdots \\ \mathbf{v}(\alpha_M)^H \end{bmatrix}, \quad \alpha_i = \frac{2\pi i}{M'} \quad (i = -M, \dots, M) \quad (14)$$

The resulting real-valued beamspace manifold is:

$$\mathbf{a}_r(\theta, \phi) = \mathbf{F}_r^H \mathbf{a}(\theta, \phi) = \sqrt{N} \cdot \mathbf{W}^H \mathbf{J}(\zeta) \mathbf{v}(\phi) \quad (15)$$

And the array output can be expressed as:

$$\mathbf{a}_r(\theta, \phi) = \begin{bmatrix} f(\zeta, \phi - \alpha_{-M}) \\ \vdots \\ f(\zeta, \phi - \alpha_0) \\ \vdots \\ f(\zeta, \phi - \alpha_M) \end{bmatrix} \quad (16)$$

Where:

$$f(\zeta, \phi) = \sqrt{\frac{N}{M'}} \left[J_0(\zeta) + 2 \sum_{m=1}^M J_m(\zeta) \cos(m\phi) \right] \in \mathbb{R} \quad (17)$$

Further details on phase-mode excitation are available in [3, 2].

After the beamspace transformation, the UCA manifold is mapped onto an M' -dimensional virtual ULA. The resulting virtual ULA is partitioned into $S = M' - M'' + 1$ overlapping M'' -dimensional subarrays for mode-space spatial smoothing [4], where $M'' \leq M' - D + 1$. To further enhance de-correlation, forward-backward averaging can be used. Unlike prior works, the proposed method retains the Bessel-function tapering.

$$R_{fb} = \frac{1}{2S} \sum_{l=1}^S Z_l^T [R + \bar{I}(R)^* \bar{I}] Z_l \quad (18)$$

where

$$Z_l = [e_l, e_{l+1}, \dots, e_{l+M''-1}] \quad (19)$$

with e_l denoting the l^{th} column of the identity matrix of size $2M + 1$. S is the number of sub-arrays, $(\cdot)^*$ denotes complex conjugate, and $\bar{I}$ is the exchange matrix.

Post-smoothing decorrelation is governed by the effective correlation coefficient [5]

$$|\rho_f| = \left| \rho \frac{\sin(S(\theta_i - \theta_j)/2) \cos \psi}{S \sin((\theta_i - \theta_j)/2)} \right| \quad (20)$$

Here, ψ is the phase angle of the complex correlation coefficient ($\rho = |\rho|e^{j\psi}$). ρ_f depends *only* on the angular separation $(\theta_i - \theta_j)$, not on individual directions, and is completely independent of inter-element spacing of array elements.

3 Dynamic Noise Variance Thresholding and Subspace Separation

Successful Direction-of-Arrival (DoA) estimation requires reliable separation of the signal and noise subspaces. After spatial smoothing, the raw pairwise signal correlations are scaled by the Dirichlet kernel. For strictly coherent sources, where the pre-smoothing correlation magnitude is $|\rho| = 1$, the effective $D \times D$ post-smoothing correlation matrix C is given by

$$C_{ij} = \begin{cases} 1, & i = j \\ \rho_f(\theta_i, \theta_j), & i \neq j \end{cases} \quad (21)$$

The smoothed signal covariance matrix is

$$\overline{R}_s = P^{1/2} C P^{1/2} \quad (22)$$

with source powers $P = \text{diag}(P_1, \dots, P_D)$.

Assuming an ill-conditioned coherent signal subspace, the inverse trace is dominated by the smallest eigenvalue:

$$\text{Tr}(\overline{R}_s^{-1}) = \sum_{i=1}^D \frac{1}{\lambda_i} \approx \frac{1}{\lambda_D(\overline{R}_s)} \quad (23)$$

$$\lambda_D(\overline{R}_s) \approx \left\{ \text{Tr}(P^{-1/2} C^{-1} P^{-1/2}) \right\}^{-1} = \left\{ \sum_{i=1}^D \frac{[C^{-1}]_{ii}}{P_i} \right\}^{-1} \quad (24)$$

Relating the total data covariance matrix R_x to the smoothed source covariance gives $\lambda_D(R_x) = \lambda_D(A_s \overline{R}_s A_s^H) + \sigma_n^2$. Since the non-zero eigenvalues of XY equal those of YX , and assuming nearly orthogonal beamspace steering vectors ($A_s^H A_s \approx M'' I$), the array gain $M'' = M' - S + 1$ factors out:

$$\lambda_D(R_x) \approx (M' - S + 1) \left\{ \sum_{i=1}^D \frac{P_i}{[C^{-1}]_{ii}} \right\} + \sigma_n^2 \quad (25)$$

For successful partitioning, this smallest signal eigenvalue must exceed the noise threshold $K\sigma_n^2$, where K is a threshold multiplier. This yields the baseline inequality for source enumeration:

$$(M' - S + 1) \left\{ \sum_{i=1}^D \frac{P_i}{[C^{-1}]_{ii}} \right\} \geq (K - 1)\sigma_n^2 \quad (26)$$

Equation (26) shows that S and K must be jointly selected so that the decorrelated signal energy exceeds the threshold margin.

The noise variance is estimated from the smallest p eigenvalues:

$$\hat{\sigma}_n^2 = \frac{1}{p} \sum_{i=M''-p+1}^{M''} \lambda_i \quad (27)$$

The dynamic signal subspace boundary is then defined as

$$\gamma = K \cdot \hat{\sigma}_n^2 \quad (28)$$

Eigenvalues satisfying $\lambda_i > \gamma$ are classified as belonging to the restored signal subspace.

4 Cramér–Rao Bound for Azimuth Estimation

To benchmark the proposed estimator, we compute the deterministic CRB [6] for azimuth under an independent-source reference model. Since the simulations report azimuth error only, the bound is derived for ϕ with $\theta = 90^\circ$ fixed.

Assuming uncorrelated sources $\mathbf{R}_s = \text{diag}(P_1, \dots, P_D)$, the azimuth Fisher information matrix is [7]:

$$\mathbf{J}_\phi = \frac{2T}{\sigma_n^2} \Re \{ (\mathbf{D}_\phi^H \mathbf{P}_\mathbf{A}^\perp \mathbf{D}_\phi) \circ \mathbf{R}_s^T \} \quad (29)$$

where $\mathbf{P}_\mathbf{A}^\perp = \mathbf{I} - \mathbf{A}(\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H$ is the orthogonal projector, and $\mathbf{D}_\phi = [\partial \mathbf{a}(\phi_1)/\partial \phi_1, \dots, \partial \mathbf{a}(\phi_D)/\partial \phi_D]$ gathers the spatial gradients. For the n th UCA sensor with position $\gamma_n = 2\pi(n-1)/N$, the response is $a_n(\theta, \phi) = \exp(jK_0 r \sin \theta \cos(\phi - \gamma_n))$, yielding the partial derivative:

$$\frac{\partial a_n(\theta, \phi)}{\partial \phi} = -jK_0 r \sin \theta \sin(\phi - \gamma_n) a_n(\theta, \phi) \quad (30)$$

The deterministic CRB matrix is $\text{CRB}(\phi) = \mathbf{J}_\phi^{-1}$, providing the azimuth RMSE lower bound for source k :

$$\text{RMSE}_{\text{CRB},k} = \frac{180}{\pi} \sqrt{[\mathbf{J}_\phi^{-1}]_{kk}} \quad (31)$$

5 Discussion

The threshold multiplier K was calibrated empirically using the noise-only covariance spectrum after phase-mode smoothing and forward-backward averaging. In compact arrays with finite snapshots, these operations introduce noise eigenvalue spreading, causing conventional AIC and MDL assumptions to fail.

Monte Carlo simulations using spatially white Gaussian noise were used to estimate the empirical cumulative distribution function (CDF) of the normalized noise eigenvalues,

$$\lambda_i / \hat{\sigma}_n^2,$$

where $\hat{\sigma}_n^2$ denotes the tail-eigenvalue, which is taken as a conservative noise variance estimate.

6 Conclusion

This work investigated the limits of a compact five-element Uniform Circular Array (UCA) for line-of-sight (LOS) recovery in coherent and adversarial RF environments with non-minimum phase channels. By combining mode-space transformation, spatial smoothing, and forward-backward averaging, the proposed UCA-RB-MUSIC framework restores covariance rank under coherent multipath conditions and enables stable subspace separation through dynamic noise-variance thresholding. Across LOS-dominant and fully scattered propagation scenarios, the method consistently improves localization stability, approaches the deterministic Cramér–Rao Bound (CRB) at appreciable operational SNR, and maintains robust LOS isolation under moderate coherent interference. These results demonstrate the practical viability of compact UCAs for sensing, electronic support, and passive radar applications in interference-dominated environments.

7 Implementation

Algorithm 1 UCA-RB-MUSIC DoA Estimation with Spatial Smoothing

Require: $\mathbf{X}_{\text{raw}} \in \mathbb{C}^{N \times T_{\text{total}}}$, r , N , f , T , K , Φ

Ensure: $\mathbf{P}_{\text{dB}}(\theta, \phi)$, $\{(\hat{\theta}_d, \hat{\phi}_d)\}_{d=1}^D$

▷ **Phase I: Initialization**

- 1: $\lambda \leftarrow c/f$, $k_0 \leftarrow 2\pi/\lambda$, $M \leftarrow \lfloor k_0 r \rfloor$, $M' \leftarrow 2M + 1$
- 2: Set beamspace dimension M'' and $L \leftarrow M' - M'' + 1$
- 3: Construct $\mathbf{W}$ with $W_{m,n} \leftarrow M'^{-1/2} e^{jm2\pi n/M'}$
- 4: $\mathbf{X} \leftarrow \Phi \odot \mathbf{X}_{\text{raw}}$, $\mathbf{P}_{\text{sum}} \leftarrow \mathbf{0}$

▷ **Phase II: Frame Processing**

- 5: **for** $k = 1, \dots, K$ **do**
- 6: $\mathbf{X}_k \leftarrow \mathbf{X}(:, (k-1)T + 1 : kT)$, $\mathbf{R}_s \leftarrow \mathbf{0}$
- 7: **for** $l = 1, \dots, L$ **do**
- 8: $\mathbf{Y}_l \leftarrow \mathbf{F}_r^{(l)} \mathbf{X}_k$
- 9: $\mathbf{R}_l \leftarrow T^{-1} \mathbf{Y}_l \mathbf{Y}_l^H$
- 10: $\mathbf{R}_s \leftarrow \mathbf{R}_s + \frac{1}{2} (\mathbf{R}_l + \mathbf{J} \mathbf{R}_l^* \mathbf{J})$
- 11: **end for**
- 12: $\mathbf{R}_s \leftarrow L^{-1} \mathbf{R}_s$
- 13: $\mathbf{R}_s \leftarrow \mathbf{V} \mathbf{\Lambda} \mathbf{V}^H$, $\lambda_1 \geq \dots \geq \lambda_{M''}$
- 14: Estimate $\hat{D}$ using the proposed noise-floor threshold
- 15: $\mathbf{E}_n \leftarrow [\mathbf{v}_{\hat{D}+1}, \dots, \mathbf{v}_{M''}]$
- 16: **for each** (θ_i, ϕ_j) **do**
- 17: $\mathbf{a}_r \leftarrow \mathbf{F}_r \mathbf{a}(\theta_i, \phi_j)$
- 18: $P_k(\theta_i, \phi_j) \leftarrow (\mathbf{a}_r^H \mathbf{E}_n \mathbf{E}_n^H \mathbf{a}_r)^{-1}$
- 19: **end for**
- 20: $\mathbf{P}_{\text{sum}} \leftarrow \mathbf{P}_{\text{sum}} + |P_k|$
- 21: **end for**

▷ **Phase III: DoA Extraction**

- 22: $\bar{\mathbf{P}} \leftarrow K^{-1} \mathbf{P}_{\text{sum}}$
 - 23: $\bar{\mathbf{P}}_{\text{dB}} \leftarrow 10 \log_{10}(\bar{\mathbf{P}} / \max \bar{\mathbf{P}})$
 - 24: $\{(\hat{\theta}_d, \hat{\phi}_d)\}_{d=1}^D \leftarrow \text{top } D \text{ regional maxima of } \bar{\mathbf{P}}_{\text{dB}}$
 - 25: **return** $\bar{\mathbf{P}}_{\text{dB}}$, $\{(\hat{\theta}_d, \hat{\phi}_d)\}_{d=1}^D$
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8 Appendices

8.1 Statistical Analysis of the Signal Eigenvalue Inequality

This appendix derives a sufficient condition for reliable signal–noise subspace separation under coherent multipath and interference. The analysis considers spatial smoothing without forward-backward averaging (FBA).

Let the smoothed signal covariance be

$$\bar{\mathbf{R}}_s = \mathbf{P}^{1/2} \mathbf{C} \mathbf{P}^{1/2}, \quad (32)$$

$$\text{Tr}(\bar{\mathbf{R}}_s^{-1}) = \sum_{i=1}^D \frac{[C^{-1}]_{ii}}{P_i} = \sum_{i=1}^D \frac{1}{\lambda_i}. \quad (33)$$

For a unit-diagonal correlation matrix, the Schur complement gives

$$[C^{-1}]_{ii} = \frac{1}{1 - R_i^2}, \quad (34)$$

where R_i^2 is the multiple correlation coefficient of source i with the remaining sources. Hence, in the strongly coherent regime,

$$\lambda_D(\bar{\mathbf{R}}_s) \approx \left(\sum_{i=1}^D \frac{1}{P_i(1 - R_i^2)} \right)^{-1}. \quad (35)$$

$$\lambda_D(\bar{\mathbf{R}}_s) \sim P_{\min}(1 - R_{\max}^2). \quad (36)$$

$$\mathbf{R}_x = \mathbf{A}_s \bar{\mathbf{R}}_s \mathbf{A}_s^H + \sigma_n^2 \mathbf{I}. \quad (37)$$

$$\mathbf{A}_s^H \mathbf{A}_s \approx (M' - S + 1) \mathbf{I}, \quad (38)$$

$$\lambda_D(\mathbf{R}_x) \approx (M' - S + 1) \lambda_D(\bar{\mathbf{R}}_s) + \sigma_n^2. \quad (39)$$

$$(M' - S + 1) \lambda_D(\bar{\mathbf{R}}_s) \geq (K - 1) \sigma_n^2. \quad (40)$$

Using the conservative approximation above yields the sufficient condition

$$(M' - S + 1) P_{\min}(1 - R_{\max}^2) \gtrsim (K - 1) \sigma_n^2 \quad (41)$$

The bound therefore captures the dependence of subspace separability on effective aperture, source power, coherence, and the detection threshold: increasing aperture or signal power improves separability, whereas increasing coherence degrades it.

8.2 Cramér–Rao Bound for DoA Azimuth Estimation

Consider an N -element sensor array receiving narrowband signals from D sources over T snapshots. The observed baseband data vector at snapshot t is modeled as

$$\mathbf{X}(t) = \mathbf{A}(\boldsymbol{\phi}) \mathbf{S}(t) + \mathbf{N}(t), \quad t = 1, 2, \dots, T \quad (42)$$

where $\boldsymbol{\phi} = [\phi_1, \phi_2, \dots, \phi_D]^T \in \mathbb{R}^{D \times 1}$ denotes the unknown azimuth angles, $\mathbf{A}(\boldsymbol{\phi}) = [\mathbf{a}(\phi_1), \mathbf{a}(\phi_2), \dots, \mathbf{a}(\phi_D)] \in \mathbb{C}^{N \times D}$ is the steering matrix, $\mathbf{S}(t) \in \mathbb{C}^{D \times 1}$ contains the deterministic source waveforms, and $\mathbf{N}(t)$ is zero-mean circularly symmetric spatially white Gaussian noise with covariance

$$\mathbb{E}[\mathbf{N}(t) \mathbf{N}^H(s)] = \sigma_n^2 \mathbf{I} \delta_{t,s}.$$

Conditioned on these parameters, the likelihood of the complete data matrix $\mathbf{X} = [\mathbf{X}(1), \dots, \mathbf{X}(T)]$ is

$$p(\mathbf{X}; \boldsymbol{\phi}, \mathbf{S}, \sigma_n^2) = \frac{1}{(\pi \sigma_n^2)^{NT}} \exp \left(-\frac{1}{\sigma_n^2} \sum_{t=1}^T \|\mathbf{X}(t) - \mathbf{A}(\boldsymbol{\phi}) \mathbf{S}(t)\|^2 \right). \quad (43)$$

$$\mathcal{L}(\boldsymbol{\phi}, \mathbf{S}, \sigma_n^2) = -NT \ln(\pi \sigma_n^2) - \frac{1}{\sigma_n^2} \sum_{t=1}^T \|\mathbf{X}(t) - \mathbf{A}(\boldsymbol{\phi})\mathbf{S}(t)\|^2. \quad (44)$$

The Fisher Information Matrix (FIM) quantifies the sensitivity of the likelihood to perturbations in the unknown parameters. Eliminating the deterministic signal parameters $\mathbf{S}(t)$ via partial differentiation, Stoica and Nehorai [6, 7] showed that the concentrated FIM for the spatial parameter vector $\boldsymbol{\phi}$ can be written using the orthogonal projector onto the array noise subspace:

$$\mathbf{P}_\mathbf{A}^\perp = \mathbf{I} - \mathbf{A}(\boldsymbol{\phi}) (\mathbf{A}^H(\boldsymbol{\phi})\mathbf{A}(\boldsymbol{\phi}))^{-1} \mathbf{A}^H(\boldsymbol{\phi}) \quad (45)$$

and the derivative matrix

$$\mathbf{D}_\phi = \left[\frac{\partial \mathbf{a}(\phi_1)}{\partial \phi_1}, \frac{\partial \mathbf{a}(\phi_2)}{\partial \phi_2}, \dots, \frac{\partial \mathbf{a}(\phi_D)}{\partial \phi_D} \right] \in \mathbb{C}^{N \times D}. \quad (46)$$

The sample source covariance across T snapshots is

$$\mathbf{R}_s = \frac{1}{T} \sum_{t=1}^T \mathbf{S}(t) \mathbf{S}^H(t). \quad (47)$$

$$\mathbf{R}_s = \text{diag}(P_1, P_2, \dots, P_D), \quad (48)$$

Using the concentrated asymptotic formulation of Stoica and Nehorai [6, 7], the FIM for joint azimuth estimation is

$$\mathbf{J}_\phi = \frac{2T}{\sigma_n^2} \Re \{ (\mathbf{D}_\phi^H \mathbf{P}_\mathbf{A}^\perp \mathbf{D}_\phi) \circ \mathbf{R}_s^T \} \quad (49)$$

where $\Re\{\cdot\}$ denotes the real-part operator and $\circ$ represents the Hadamard product.

$$\gamma_n = \frac{2\pi(n-1)}{N}, \quad n = 1, 2, \dots, N. \quad (50)$$

The steering response of the n^{th} element to a source arriving from azimuth ϕ_k is

$$a_n(\phi_k) = \exp(jK_0 r \cos(\phi_k - \gamma_n)), \quad (51)$$

$$\frac{\partial a_n(\phi_k)}{\partial \phi_k} = -jK_0 r \sin(\phi_k - \gamma_n) a_n(\phi_k). \quad (52)$$

$$\frac{\partial \mathbf{a}(\phi_k)}{\partial \phi_k} = \begin{bmatrix} -jK_0 r \sin(\phi_k - \gamma_1) a_1(\phi_k) \\ -jK_0 r \sin(\phi_k - \gamma_2) a_2(\phi_k) \\ \vdots \\ -jK_0 r \sin(\phi_k - \gamma_N) a_N(\phi_k) \end{bmatrix}. \quad (53)$$

$$\text{CRB}(\boldsymbol{\phi}) = \mathbf{J}_\phi^{-1}. \quad (54)$$

$$\text{var}(\hat{\phi}_k) \geq [\mathbf{J}_\phi^{-1}]_{kk}, \quad k = 1, 2, \dots, D. \quad (55)$$

$$\text{RMSE}_{\text{CRB}}(\phi_{\text{LOS}}) = \frac{180}{\pi} \sqrt{[\mathbf{J}_\phi^{-1}]_{\text{LOS}, \text{LOS}}}. \quad (56)$$

Because spatial smoothing sacrifices effective array aperture in order to restore covariance rank, comparing the measured RMSE of the smoothed estimator against this independent-source CRB provides a direct measure of the asymptotic efficiency and residual decorrelation penalties of the compact five-element sub-array architecture.

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